

Prelab 5: Ohm's Law

March 7, 2026

1. A resistor with a green-blue-orange-gold color code has resistance:

$$R = 56 \cdot 1 \text{ k}\Omega \pm 5\% = \mathbf{56 \text{ k}\Omega \pm 2.8 \text{ k}\Omega}.$$



2. If R_1, R_2, R_3 are the resistances of three resistors in parallel, their equivalent resistance is:

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \rightarrow R_{\text{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)^{-1}.$$

If they are instead connected in series, then:

$$R_{\text{eq}} = R_1 + R_2 + R_3.$$

In the PhET Lab, I used three resistors with resistances $R_1 = 10 \Omega$, $R_2 = 15 \Omega$, $R_3 = 20 \Omega$.

- In parallel, the voltmeter measured a potential difference of 12.00 V across the resistors, and the ammeter measured a current of 2.60 A flowing through the circuit. Thus, the equivalent resistance of the parallel combination is:

$$R_{\text{eq}} = \frac{\Delta V}{i} = 4.61 \Omega.$$

Using the equation from above, the calculated equivalent resistance of the parallel combination is:

$$R_{\text{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)^{-1} = \left(\frac{1}{10 \Omega} + \frac{1}{15 \Omega} + \frac{1}{20 \Omega} \right)^{-1} = 4.62 \Omega.$$

This is consistent with the measured value.

- In series, the voltmeter measured $\Delta V = 12.00 \text{ V}$ across the resistors, and the ammeter measured $i = 0.27 \text{ A}$:

$$R_{\text{eq}} = \frac{\Delta V}{i} = 44.44 \Omega.$$

Using the equation from earlier:

$$R_{\text{eq}} = R_1 + R_2 + R_3 = 10 \Omega + 15 \Omega + 20 \Omega = 45 \Omega.$$

This is consistent with the measured value.

3. Using Kirchhoff's Loop Rule, the current through the circuit is:

$$+V_{\text{in}} - iR_1 - iR_2 = 0$$

$$i(R_1 + R_2) = V_{\text{in}} \rightarrow i = \frac{V_{\text{in}}}{R_1 + R_2}.$$

The potential difference across R_1 , $V_{\text{out}}(R_1)$, is:

$$V_{\text{out}}(R_1) = iR_1 = \frac{V_{\text{in}} R_1}{R_1 + R_2}.$$

